

Team Members

Aryana	Legal Analysis & RASE Interpretation
Othman	Documentation & Prompt Engineering
Samih	Technical Development & Software Integration

1. Project Objective

The objective of this project is to investigate how legal building regulations can be transformed into structured, machine-readable representations using the **RASE methodology** (Requirement, Applicability, Selection, Exception). The project evaluates whether Large Language Models (LLMs) such as ChatGPT, Claude, Copilot, or Gemini can reproduce human-created RASE annotations. The selected regulation is **§47 Abs. 1 of the German Musterbauordnung (MBO)**, which defines requirements for habitable rooms (Aufenthaltsräume).

2. Project Structure

1 Phase 1 – Manual RASE Analysis

Objective: Create a correct human-generated RASE annotation as the reference model for the entire project.

Activities	Deliverables
• Study and interpret §47 MBO	• Manual RASE analysis
• Identify Requirements, Applicability conditions, Selections, and Exceptions	• Final HTML mark-up
• Discuss ambiguous legal wording	• Logical rule interpretation
• Apply RASE mark-up using the ACCORD Rule Formalization Tool	• Documentation of legal reasoning
• Generate the final HTML annotation	
• Convert legal text into machine-readable logical rules	

2 Phase 2 – LLM Prompt Engineering

Objective: Determine whether a Large Language Model can reproduce the same RASE annotation generated by humans.

Activities	Deliverables
• Develop an initial RASE prompt	• Prompt Versions 1–3
• Submit the legal clause to multiple LLMs	• Comparison tables
• Analyse the generated annotations	• Error analysis
• Compare AI-generated mark-up with the manual reference	• Final optimized prompt
• Refine the prompt through multiple iterations	
• Measure improvements after each refinement	

Objective: Develop a software prototype that automates the interaction between legal text, LLMs, and RASE analysis.

Activities	Deliverables
• Create a web-based interface	• JavaScript application
• Allow users to input legal clauses	• User interface
• Send legal text and prompts to an LLM	• LLM integration
• Display generated RASE annotations	• Annotation visualization
• Visualize extracted RASE elements	• Demonstration system
• Compare AI-generated results with human-generated annotations	

3. Team Responsibilities

Aryana	Othman	Samih
• Legal interpretation of §47	• Prompt engineering	• Technical implementation
• Identification of RASE elements	• Prompt refinement	• HTML validation
• Validation of manual annotations	• Documentation and reporting	• Rule formalization
• Documentation of annotation decisions	• Comparison and evaluation of AI outputs	• Development of the JavaScript application
		• Integration of AI-generated outputs

4. Expected Outcomes

01	Produce a complete and correct RASE annotation of §47 MBO.
02	Create a machine-readable representation of the regulation.
03	Evaluate the ability of LLMs to reproduce human-generated annotations.
04	Develop a proof-of-concept application demonstrating AI-assisted rule formalization.
05	Assess the strengths and limitations of using LLMs in regulatory compliance.

5. Future Research

Future work will focus on testing additional building regulations and evaluating whether LLMs can consistently generate accurate RASE annotations across different legal domains. The long-term goal is to explore how AI can support **semi-automated rule formalization** while maintaining human oversight and validation.